

POCKETSMART AI: PROPOSED TECHNICAL SOLUTION

1. Solution Overview

PocketSmart AI is a lightweight, responsive, and intelligent multi-category financial planning system. Powered by Python 3.12, FastAPI, Bootstrap 5, and Google Gemini AI, the application converts high-level budgetary constraints into structured, purchase-ready itemized plans.

2. Key Pillars of the Proposed Solution

2.1 Multi-Category Modular Architecture

The application avoids one-size-fits-all budgeting by providing three dedicated planning engines:

1. **Home Interior Planner:** Tailored for domestic renovation, managing room counts, illumination fixtures, and furniture pieces.
2. **Party & Event Planner:** Calibrated for social gatherings, corporate events, and celebrations based on headcounts and meal requirements.
3. **Jewelry & Lifestyle Planner:** Specialized for festive and wedding purchases incorporating multimodal visual inputs.

2.2 Intelligent Hybrid Engine (AI + Algorithmic Fallback)

- **Primary AI Engine:** Uses Google Gemini generative models to produce dynamic context-aware shopping suggestions and room layouts.
- **Algorithmic Fallback Engine:** In the event of network latency, API rate limits, or offline conditions, the system automatically activates a heuristic rules engine, calculating exact Indian market allocations without failing or returning error codes.

2.3 Native E-Commerce Search Link Generation

Every recommended item is programmatically transformed into URL-encoded search links targeting prominent e-commerce platforms:

- *Home Planner*: Direct links to Amazon India, Flipkart, and IKEA.
- *Party Planner*: Direct links to Swiggy Catering, Zomato, and OYO event spaces.
- *Jewelry Planner*: Direct links to Amazon, Tanishq, and Myntra.

2.4 Lightweight Token Security & JSON Persistence

- Uses OAuth2 with JSON Web Tokens (JWT) and Bcrypt encryption for user credentials.
- Employs localized `users.json` dynamic persistence, ensuring user accounts persist seamlessly without heavy database overhead.